Modeling the spread of vaccination behavior and infection on survey-derived networks

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MOTIVATION

- Dual Contagion: Pathogens spread through physical proximity, while behaviors (like vaccination) may spread through social influence via technology and spatial proximity.
- Network Influence: Spatial and social interactions can be summarized using a **network**. Network structure can dictate the speed and magnitude of both disease and social contagion.
- The U.S. Gap: There is limited data on higher-order network metrics (beyond simple degree) for the large, heterogeneous U.S. population.
- Epidemiological Risk: When groups of similar individuals are highly connected, infections can circulate within them, and protection (like vaccination) may not reach them as easily. Understanding these patterns helps us better assess risk and design interventions.

DATA

- RAND American Life Panel (February 2022) survey of 2,145 U.S. participants.
- Ego-Network Construction:
  - Nodes: 2,145 egos (participants) and up to 15 alters (contacts) per person.
  - Edges: Differentiated by contact type (Social vs. Spatial) and frequency.
  - Demographics: Includes age, gender, race, urbanicity, and vaccination habits.

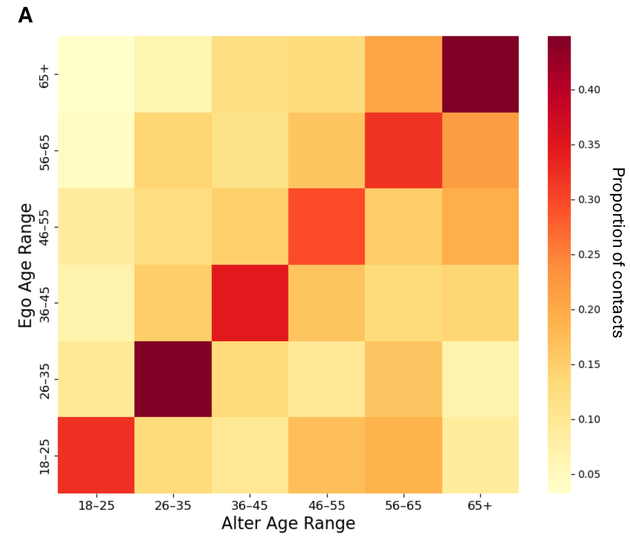
APPROACH

- Higher-Order Metrics: Calculate clustering and assortativity by age, gender, and vaccination status.
- Network Modeling: Use Exponential Random Graph Models (ERGMs) to make more complete social and spatial networks based on survey-derived patterns.
- Social Contagion Model: Test three behavioral mechanisms for vaccination uptake:
  - Simple Contagion: Requires only one exposure to spread information.
  - Complex Contagion: Requires exposure to multiple individuals to spread information
  - Historical Habit: Persistence of previous years' behavior.
- Disease Model: Run a standard SIR (Susceptible-Infected-Recovered) model on the spatial network.
  - Transmissibility rate: 0.05 to 0.5 to reflect varying respiratory pathogens.
  - Recovery rate: 5–7 days (standard influenza parameters).

TAKEAWAYS

- Our dataset is unique in that it provides participant data as well as data on their social and spatial contacts.
  - This allows us to get a bigger picture of how behavior and disease spreads through a population that is similar to the U.S. population.
- With our analysis we can:
  - Determine how infection clusters are a result of physical contact patterns and behavioral clusters.
  - Generalize which demographic groups are at a greater risk of infection based on their network structures.
- So far, we can see that:
  - Two contacts of an individual are likely to also be in contact with each other and this consistent of demographic groups. There is variation in this metric across groups.
  - People tend to have contact within similar age, gender, and vaccination status groups.
    - This creates pockets of susceptibility to disease for similarly vaccinated groups.
  - Altered behavioral patterns yield different vaccination results.

Who interacts with whom?



How interconnected are groups?

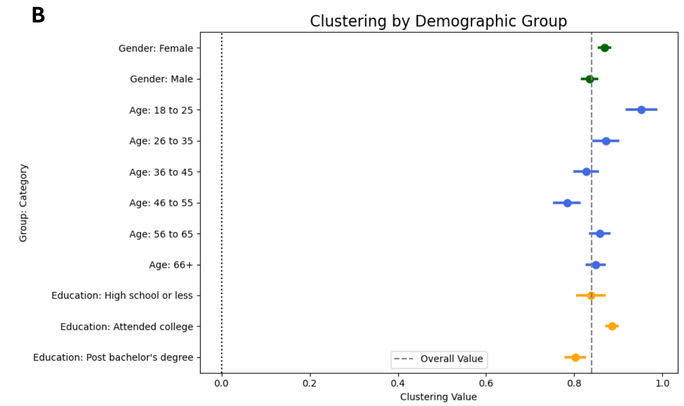
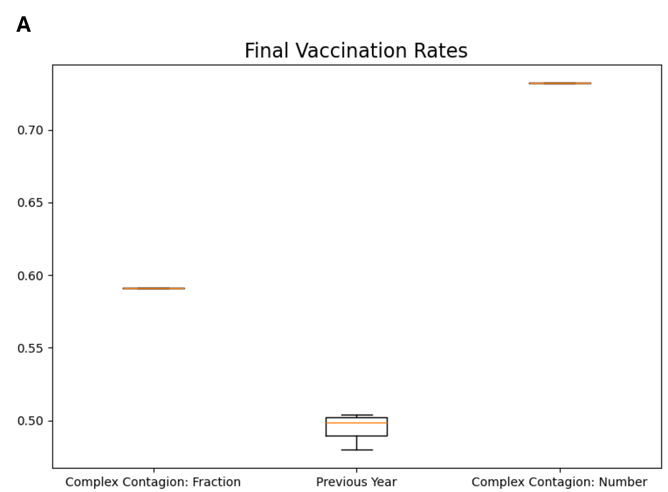


Fig 1. a. Assortativity of contacts by age groups. The positive correlation between groups show that individuals tend to have more contact within their age group than with other age groups. b. Clustering of contacts by demographic groups gender, age, and education. The dotted line represents zero clustering, the dashed line represents average clustering across all participants. Clustering is quite similar across demographics with the most variation across age groups with younger groups having higher than average clustering.

Simulating vaccination behavior with our model



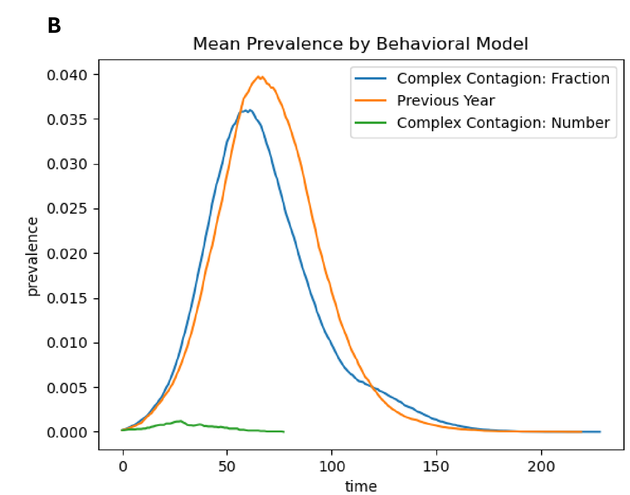


Fig 2. a. Final vaccination rates of simulations of each model. The highest vaccination rate is with the complex contagion model where individuals are vaccinated once at least 5 of their neighbors are vaccinated, the next is with the complex contagion model where an individual is vaccinated once 70% of their neighbors are vaccinated, and the lowest vaccination rate is where the vaccination status of individuals is determined by their vaccination status from the previous year. Each simulation started with a 45-50% vaccination rate against a disease with a basic reproduction number of 5. b. The mean prevalence of simulations for each model. The highest peak prevalence is with vaccination determined by the previous year, then with complex contagion by proportion, with the lowest prevalence occurring with the complex contagion by number of neighbors.